

Tempestology Part 2 with Matt Lanza & Dr. Kim Wood

Ologies Podcast

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Oh hey, it's still your ex-boyfriend who pretended he didn't know how to wash the cast iron right, so he just didn't, Alie Ward, and this is Part 2 of a smooth sail through rough weather. This is *Ologies*; this is hurricanes, typhoons, or cyclones. We describe the difference in Part 1. And in Part 1, we met Dr. Kim Wood, who's an associate professor in the Department of Hydrology and Atmospheric Sciences at the University of Arizona. And also, Matt Lanza, a Rutgers-trained forecast meteorologist and a writer and editor for Houston's highly respected, deeply beloved meteorological outlet, Space City Weather. He's also a co-founder of the website, the Eyewall.

We covered how fast hurricanes go, with the threshold for a Category 1 being 74 miles an hour. And Kim blew open my mind, explaining that if you stuck your hand outside a moving car window, think of that force. A Category 5, as covered in Part 1, is at least 157 miles per hour, or 252 kilometers an hour or higher. Catastrophic indeed. So, when you're thinking about these wind speeds, like... [croaky groan] I can't imagine.

Okay. But before we dive into this Part 2, thank you to patrons of the show who make *Ologies* possible. You can join for a dollar or more a month and send in questions for the ologists via Patreon.com/Ologies. Thank you to everyone out there in merch from OlogiesMerch.com; you can get hats and shirts and totes, and you all can find each other in the wild. And thank you to everyone who leaves a review for me to read, which helps us so much, keeps us up in the charts. I read all of them like a little, sad, lonely elf alone in a dark room, including this review this week from Quizeeny, who wrote that they enjoy *Ologies* so much that:

Maybe you're stalking me and created this podcast just for me.

Quizeeny, maybe you're me from another dimension. Wouldn't that be nice?

Okay. If you need 400 or more *Ologies* episodes, we got 'em. Just go to Ologies.com, they're all neatly sorted into topic, so you can find as many as you like. Also, if you do not like my sometimes-filthy mouth, we have a spinoff show called *Smologies*, and you can subscribe wherever you get your podcasts. It's for all ages, kid-safe, shorter versions, they're at the link in the show notes or wherever you get your podcasts, *Smologies*.

Okie dokes. Let's dive into Part 2 of Tempestology. It's all about how to prepare for a hurricane or really any disaster that comes your way, the very latest on weather and climate science funding, where you should retire, what's Waffles got to do with it, and what happens if you forge a hurricane map on international TV. We'll hear from professional disasterologist and icon, Dr. Sam Montano, about the current climate of things and some of her reflections on equity and disaster relief, and what you should do now in case something hits ya. So, batten down the hatches, let's barrel headfirst into cyclonic systems with tempestologists Dr. Kim Wood and Matt Lanza.

Alie: When it comes to peril in general for animals, ecosystems, people, is it the flooding or is it the abrasive quality of these sustained winds? What is it that is so, I guess, scary, that requires so much foresight?

Kim: There's a couple of complicating factors here. So, you're going to have widespread trees fallen on roads, power lines down, all kinds of infrastructural elements that we rely on for day-to-day life have been thwacked. ["Oof."] Often, these places, you know, waters may have receded, but then

you've got all these damaged, soaked places, mold issues, all kinds of things. So, the impacts are felt far and wide for a long time. So, the water can be bad from that perspective, but like, kind of in the moment. The power of water cannot be underestimated because one foot of moving water can knock over a person. So, you know, just add feet and you're adding impact.

You've got the winds on top of that. So, it's hard to, kind of, differentiate the water from the wind in that context, especially if you also get hit with a tornado warning at the same time, because getting away from water, you go up in a house; getting away from a tornado, you go down in a house. Well, what do you do when you have both? That's one reason why evacuation can be encouraged for some of those most vulnerable areas, but not everyone can evacuate. But then you have a situation like Hurricane Florence in 2018, where it dumps a lot of rain.

Alie: Oh my god.

Kim: Yeah. So, it really is a multifaceted hazard, highly dependent on where you're located. Harvey demonstrated that with its ridiculous five feet of rain record. I can't wrap my head around five feet of rain. *[laughs]*

Alie: I can't. I can't even... I mean, I live in LA, so I can't wrap my head around a couple inches. It's like, everyone cancels their plans.

Kim: Yeah, we got two inches of rain last night, and we're like, "What? Where'd all this come from?"

Alie: It's like, Southwest United States, roll call. *[Kim laughs]* Present! *"It's dry. It's really dry."*

Aside: But in August of 2023, LA did experience the torrential remnants of Hurricane Hilary on the same day as a 5.1 magnitude earthquake. I was there for both of them. It was a real apocalyptic situation. Also, this January, LA erupted into flames. So, you know...

But Kim says that the risks are many-faceted, from winds to rains, to the potential of tornadoes. But what about just stuff?

Alie: Andrew McVay wanted to know: Is there a hurricane rating defined by the average amount or mass of debris being carried through the air? Is that ever taken into account when warning the public, like, if it's going to hit an urban area, or obviously, with Katrina in New Orleans, the basin of the city had a huge impact on the infrastructure of the levees and stuff. Is that ever taken into account? Is there a warning system that might develop from that?

Matt: There's one warning that gets issued by local National Weather Service offices when you have a particularly strong hurricane coming short, and it's called an extreme wind warning. They'll post that when the eye wall of a storm, where the strongest winds are, starts to move into a particular place. It doesn't have to be urban; it can be rural, suburban, whatever. But whenever those winds are like 120, 30, 40 miles an hour, you're going to get that warning. That's basically the moment where you have to say you've got to be in the safest place humanly possible, and you just stay there until it's over.

Aside: But before we get to the safest place to be, what happens to all of the nonhumans caught in these storms?

Alie: SustainableSirenian, again, said: What happens to marine animals during hurricanes? After Milton, a hunter found a sea turtle in a forest inland. Jessica Randolph wants to know if *Sharknado* could actually happen. *[Matt laughs]* But, you know, probably not on that scale. Do you ever have to work with ecologists about, like, there's a completely different population of salamander over here because it just got swept up and dropped?

Matt: Yeah, it's really kind of fascinating when you deal with this. So, one thing that's very common, this is extremely common. What will happen is you get, like, a really strong hurricane that goes through, like, say, the Caribbean. You've got a lot of exotic birds down there, and they'll get caught in the middle of the storm, they'll get caught in the eye of the hurricane. You can even see it sometimes on radar. If you look on radar at a hurricane, you'll see what we call ground clutter. But it's actually biological... What we classify as biological material because it's birds, it's bugs, it's all this different stuff, and they get caught in the eye! So, what happens is, you know, let's say hypothetically, a storm goes through the Caribbean, comes up through Florida into the southeast, and all of a sudden, like, in North Carolina, you'll start getting reports about these ridiculous birds that they've never seen before that have just shown up and it's because the hurricane left them there.

So, you know, it's a fascinating thing. Like, I can imagine the sea turtle situation is either because the surge brought the turtle so far inland that, you know, something like that, or it just got picked up because the winds were so strong. Milton was a huge storm. You know, you could get a mini scale *Sharknado* if you have, like, a strong enough tornado over water that's able to pick something up into the air. The reality is that people think about *Twister*, the original *Twister*, and you know, the scene where the cow goes across and the cow goes across again. [*sounds of stormy chaos, "Cow! I gotta go, Julia, we got cows!"*] Theoretically, sure, that could happen. Does it usually happen? No. It's kind of crazy. But yeah, I mean, theoretically, something like that can happen on a small scale.

Hurricanes reshape the ecology, they reshape the topography, they reshape the landscape. They're just extremely powerful forces of nature that have been on Earth since the beginning of time. They're part of the system of how things work and what, kind of, frankly, makes the Earth awesome to some extent. You know, if you can look past the life and property element of it, you know, obviously, it's tragic to see that. But in the grand scheme of things, it's part of the world and always has been.

Aside: So, it's part of the world. But what if you don't want a front row seat?

Alie: Well, you know, so many populations live near a coast; that's where the water and the fish and the trade is. But when you are watching, let's say, *House Hunters International*, are you seeing people looking for vacation homes on the beach, and are you thinking, "No way. That is way too close to the water. You are in the path of hurricanes." Are you screaming at your TV? You're nodding yes.

Kim: Yeah, yeah. So, I try to meet people where they are. You know, if I'm having a conversation with somebody about something that they clearly really care about, and I have information that might change their perspective on what they're thinking about, such as retiring on a beach, I try to highlight, like, these are the things that you want to think about.

I shouldn't throw them under the bus, but my parents were thinking about doing something like that. [*laughs*] And I was pretty emphatic about the things they would need to consider if they wanted to live in a place like that. And they ended up retiring to Phoenix. [*laughs*]

Alie: That's good. That's good.

Kim: So, a different set of problems.

Alie: Different side of global warming, but at least they're close by.

Kim: Yeah. But yeah, it's tough because I understand the call of the water. Like, there's something special about having easy access to a beach. But everyone needs to assess their own personal level of risk and what they're willing to take, you know, based on their financial resources, physical resources, time resources, that sort of thing. And so, when people are thinking about building something

really cool in a risky place, it's like, if you're going to do that, think about these, you know, unlikely, but increasingly likely risky scenarios so that you go in with both eyes wide open, and if something were to happen, hopefully you've got plans in place so that you can get out of there alive, because that's what matters in the end.

Choosing a place to live is hard, packing up all your stuff is hard. You know, I never want my advice to be, "Oh, just move," because it just doesn't account for the lives people are actually living. So, when you live in a place that has a risk of hurricane impacts—and I say hurricane, but, you know, in general, tropical cyclone impacts—the things to be thinking about are who do you pay attention to? Identify a couple trusted sources. Yeah, you can check the National Hurricane Center page on a daily basis for updates. That's great. But it's nice when someone translates that, especially the cone of uncertainty, into something that is relevant to where you live, especially from a timing perspective. The watches and warnings start coming up 48 hours out, and so we want people to have taken action before those winds get to them. That's why things will often be listed as a warning, even though the center of storms is still pretty far away is because the center is the worst part. You want to be prepared before the worst part gets to you. [*"Duly noted."*]

So, take pictures of everything in your house. Have your insurance documents ready—if possible, because it depends on people's ability to prepare ahead of time—have your prescription medications stocked, think about ways to partner up with other people who may live near you, maybe you could carpool, know if you're in a flood risk zone. I'm listing all these things off, and that's a lot of responsibility placed right on the person who's like, "I don't even know what a hurricane is."

Aside: What about the walls around you, and the roof above you, and the water rising at your feet?

Alie: Jake and Ellie, first-time question-askers, wanted to know: How do people hurricane-proof their land or house? Like, how can people prepare? Was on a lot of people's minds. Vanessa Adams wanted to know: Is the likelihood of a hurricane hitting you inversely related to the amount of preparation you make for it? Signed, a chronic overthinker. [*Matt laughs*] Better prepared and have it not happen. But, you know, as we're going into this season, anything people can do... Is it sandbags? Is it emergency kits?

Matt: Yeah, the first thing that I think a lot of people don't necessarily think about is where do you live? You live in a place, but what's your risk? What is your actual risk there? Do you live in a floodplain? Which bayou, which river, which creek drains your land? So, do you have to worry about flooding from that thing? Understand the history of a place that you live; talk to your neighbors that have maybe lived there for a long time. So, understanding what your neighborhood is like and what people have been through, I think, is really important.

Everybody that lives in a hurricane-prone area should be prepared with a kit. You can go online, go to your local municipal emergency management agency, county, state, local, whatever. They will always have a hurricane checklist. What do you need to do out of a hurricane? And it'll tell you. I mean, it will lay it out for you, what you should have, what should be in your kit, what do you need to think about. I always tell people, you know, aside from all that, pay really close attention to, you know, if you have pets, what are you going to do with the pets if you have to evacuate? If you have to evacuate, you've got to make a really quick move— and you know, this can go just beyond hurricanes, this can go to wildfires and other catastrophes as well—what are you going to do about childcare? What do you do if your partner's at work? How are you going to do this stuff? What route are you going to take to get out? All these different things.

You should be mapping all this out ahead of the season so that when the trigger gets pulled and it's time to do things, you've already got that plan drafted and ready to go. You know what you have to do. I think when I first moved to Texas, one of the things I did was I mapped out an evacuation route from my house to, like, Dallas. But it was on all county roads because I had heard the horror stories during Rita of, you know, more people died during Hurricane Rita in the evacuation of Houston than in the actual storm.

Alie: [*moans*] Oh my god!

Matt: Because it was just a mass exodus, nobody was prepared for it. It's a lot different today; there are plans now, there's a lot more preparation, a lot more rules. It should be less chaotic today than it was, you know, 20 years ago. But these are all different things you've got to think about. It's just about being and understanding your risk about the place that you live. And it's not always made easy for people.

Aside: And I didn't know this, but via the 2006 paper titled "Deaths Related to Hurricane Rita and Mass Evacuation," 2.5 million evacuees along the Texas gulf coast fled before this Category 5 hurricane, Rita, which killed 111 people. But Rita's path narrowly missed some major city centers, so a lot of the deaths were caused from trying to get away from it ahead of time; overheating in gridlocked traffic, turning the AC off to save fuel in the car; dehydration, people weren't drinking water because they wanted to avoid restroom stops and trying to conserve food and whatever supplies they had. So, over half of the deaths from Hurricane Rita were people found in their vehicles. Dozens more were nursing home patients who perished in a bus fire related to the excessive heat and their onboard oxygen tanks, which is awful.

Now, Hurricane Katrina, whose 20-year anniversary is coming up on August 29th of this year, was also a Category 5, also in 2005. But Katrina's death toll was over 1,800, not including the people who died later by suicide and stress-related health conditions. Now, the hurricane passed, Katrina passed, but the catastrophic loss of life was due to over 50 breaches of these poorly constructed levees along the canals of the city. And these breaches hit lower-income areas like the Lower 9th Ward and St. Bernard Parish the worst. In a 2017 article titled "Hurricanes hit the poor the hardest," by a policy nonprofit institute, Brookings, explains that "Lower-income Americans are more likely to live in neighborhoods or buildings more susceptible to storm shocks. Substandard infrastructure in affordable housing units" pose greater threats to survival. They're built like shit, essentially. And evacuation without transportation, or even unexpected fuel and supply costs, can mean the difference between life and death. And that was the case in Katrina.

This weekend, I watched the five-part documentary *Hurricane Katrina: Race Against Time*, it was directed by Traci A. Curry, it was executive-produced by *Sinners* and *Black Panther* director Ryan Coogler. This documentary details the timeline of the storm with all this old footage, but it also illuminates that Katrina's true horror is this systemic neglect to protect vulnerable people. It is impossible to ingest this series or reflect on Katrina without tasting everything that came before and has come after that storm.

I knew that the levee breaks hit minority and poor communities the hardest, but I watched all five parts of this documentary until 3:00 in the morning, all in one sitting, and you're just left wondering, like, why the fuck didn't more people do something? How could they let people suffer and die in broad daylight like that? Where was the drinking water? Where were the boats and the buses and the safety? How did anyone get away with this? How could people far away have helped more? And how can they call people in their own city, refugees? One effect of this documentary, it's on Hulu, is that Katrina's 20th anniversary lines up side by side to today's families torn apart by ICE agents, a textbook genocide in Gaza, LGBTQ+ people stripped of rights and dignity, even

military pensions, reproductive rights blocked, voter disenfranchisement, like, the authoritarian seizing of the judicial system, and realizing that help is not coming as expected. It's like, citizens on the ground, helping each other the best they can to survive, and just hoping that the people who are paid to help might be ashamed enough to actually step in. But *Hurricane Katrina: Race Against Time*, definitely worth watching.

So, let's go back real quick. We're going to go to our 2019 Disasterology episode with Dr. Sam Montano, who was so moved by Katrina that she moved to New Orleans to study disaster management and made it her career. So, here is an excerpt from that 2019 episode with Dr. Montano.

[clip from Disasterology episode]

Aside: *Going back to recovery, Samantha, remember, spent undergrad in Louisiana after Katrina and the floods. And this was a massive catastrophe. Again, 80 % of the city flooded. Human bodies floated in the floodwaters for days at a time. Infrastructure was out all over. People left and never came back. So, for this episode, I went back and looked up some AP photos of the direct aftermath, and I literally cried over my laptop and then had nightmares. The scale of this tragedy was unthinkable. So, what was her experience like there?*

Dr. M: *So, when I lived in New Orleans, I lived there for four years. I was going to college. So, in some ways, I was mostly living on campus in Uptown, New Orleans. You know, you could walk around outside and not really know that Katrina had happened in the past few years. There were some signs here and there, but for the most part, things looked "normal." But because of the organizations I worked with and the other things that I did, I was regularly spending time in all different neighborhoods throughout the entire city.*

And, you know, when you live in a place that is going through a recovery, especially of the size of Katrina's recovery, it affects kind of every aspect of your day from certain roads being closed down because they're still doing construction on those roads, or fixing the sewer lines for the first time since the storm, like years later, or, you know, trash and recycling not being back, or it being two to three years before the street car starts running again. Every different aspect of the city had to be rebuilt, and so you're operating within this space that is not operating at its full capacity. So, that kind of eats at you, that affects your daily life. Even I, who was very much still removed from that—like, I myself was not going through a recovery, I myself was living in a place that was recovered—and even then, when I left New Orleans, I felt the difference of moving to Fargo and being in a community that was all put together and operating the way you expect a community to operate.

So yeah, it definitely eats at you. We also see that in the research, in terms of, you know, people's mental health and the way that stress manifests during recovery. We see an increase in domestic violence, an increase in suicides during recovery among people who are going through that process, and yeah, it's extremely, extremely difficult to go through, particularly as a survivor of that disaster.

Alie: *Does it ever affect you to see how people respond to certain disasters as opposed to others? Or how policymakers or political officials will respond to certain communities affected or certain disasters? Is that ever something that might get your goat?*

Dr. M: *Yeah, I'm mad, like all the time. [laughs]*

Alie: *Okay, that's what I was thinking.*

Dr. M: *Just constantly mad.*

Aside: *Just a side note. So, we recorded this in late April; the world had just watched France's structural jewel, Notre Dame Cathedral, burn in part to cinders and collapse. Over one billion dollars in donations poured out of pockets, and the church reported that most of it was from small personal donations. So, okay. They raised a billion dollars very quickly, fine. It's Europe. There are so many countries that would chip in.*

But I just read a recent Travel and Leisure article that noted an estimated 90% of the donations didn't come from Europe; they came from the US. So, what about Puerto Rico, hobbled by Hurricane Maria? They got less help from the US federal government than Texas and Florida did for Hurricanes Harvey and Irma. And as for personal donations, folks gave about \$32 million to Puerto Rico as compared to that one billion for a Paris church. Hurricane Maria's death toll in Puerto Rico, which is a US territory, is estimated at 3,057 people. The death toll for the Notre Dame fire? Zero. Huge financial discrepancies there.

Dr. M: *Certainly, everything about disasters is injustice manifesting; who is affected most directly by disasters, which communities are affected and what ways are affected, their ability to prepare for disasters, their ability to mitigate disasters, their ability to recover, their ability to literally survive disasters, all of this is tied to policies that are shaped by race, class, gender. And all of those inequalities come out in the middle of a disaster. You know, those inequalities exist in all four phases, but of course, it's during the response that they are kind of most visible and in everyone's face.*

Alie: *Mm-hm. Kanye West sometimes just calling them out on a telethon. ["George Bush doesn't care about Black people."] Back, vintage Kanye West, that is.*

Aside: *The golden days of Yeezy, before red hats and proclamations that slavery is a choice.*

And yeah, this episode with Dr. Montano was recorded in 2019. And in those six years, Kanye or Ye has only spiraled into absolutely unhinged Neo-Nazi behavior. It's just yet another upside-down horror of the last few years.

Now, in a few minutes, we're going to hear again, another update from Dr. Sam Montano on what to do. But yeah, current-day disasters, Kim said...

Kim: So many folks who live in these areas are already barely getting by. So, to assume that they have enough resources to put gas in their car, to have a car, you know, you've got folks who may use wheelchairs, you have folks with pets, you have folks with medical conditions, children, elderly relatives. There's all these different factors that folks need to consider for their own personal risk and risk management. So, when it comes to our agencies, whether it be at the local to federal level, giving us advance warning, the NHC works super hard to make sure information is getting out as best they can, as far out in advance as they can. And right now, current decision trajectories are actually hindering us in being able to do that.

Aside: Matt echoes that.

Matt: I always think of the example, like, the single mom that's working three jobs, that's got two kids, doesn't have time to go through all these things. So, you know, I think it would be nice if communities had, like, almost a welcome-to-the-town packet, here's what you need to know. And it would include things like that, because, you know, all the time, we're always talking about how

great a place is to live. But the reality is that there's always going to be something. There's always some threat; it's a blizzard, it's a hurricane, it's a wildfire, it's an earthquake, whatever it is.

Alie: Do insurance companies still call it an act of God? Or is it an act of nature? I'll have to look that up.

Matt: Yeah. But you bring up an important topic because insurance industry are the ones that are starting to... They're the market signal that climate change is a problem because everybody's insurance rates are going up, and it's not just because of wildfires in California and Colorado. It's because of hailstorms in Texas, it's because of hurricanes in Texas and Florida and the Carolinas, it's because of flooding. And there are now a lot of insurers that are no longer insuring in hurricane-prone areas. They're out, they're done because they know it's too much risk. You talk about money and what gets Americans, you know, motivated seems to be money, and that's going to be a thing.

Alie: Now, what about trying to stop a hurricane? What kind of measures, historically or in the future... Have people tried to throw salt up there or anything? What could we do?

Kim: Uhhh... [scoffs] Unfortunately, when it comes to trying to control something like a hurricane, it is massive. It is releasing so much energy, and it is already interacting with the area around it in, I would say, organized, chaotic ways. But the stuff that's happening in the storm can be occurring on really small scales. Like, if you look out and you see a single thunderstorm cloud, and then you think about all the processes happening within that, that's all happening inside of a hurricane as well.

Alie: Oh, boy!

Kim: And so, when you think about, how could humanity directly affect a hurricane? We can't. No, we can't nuke them. No, we can't drag icebergs under them. No, no, no. We are indirectly affecting them by warming the climate, but we are not capable of directly affecting something the size of a hurricane.

Aside: So, the prevailing advice from professional tempestologists is to look to your local meteorologists, who will have the best handle on evacuation advice and what areas near you may be hit the hardest. Remember that these things are hard to predict weeks and sometimes even days out, and hurricanes can ping-pong around and change course that surprises even meteorologists.

So, since Part 1 of this episode went up last week, and part of the reason we spaced out this jumbo episode into two servings is because, kind of like an approaching storm about to wallop your whole existence, there have been minute-by-minute updates. So, according to the AP News:

Since January, when Trump took office, the newly formed Department of Government Efficiency gutted NOAA and the National Weather Service, which are key for the nation's daily weather forecasts, severe storm warnings, climate monitoring, and more. Hundreds of NOAA forecasters and other employees have been cut, and National Weather Service offices around the country have had a number of vacancies.

The AP reported. Meteorologists have been like, "That's one way to shoot yourself in the crotch: during increasing numbers of climate emergencies, to stop funding them. But as of this past Friday, August 6, 2025, apparently, the government is looking to hire back 450 meteorologists, hydrologists, and radar technicians.

Now, many who took what amounted to a forced retirement or buyout are not coming back. We learned last week that those were usually the scientists closest to retirement and with the most experience. Working under an administration who's looking to, like, slash mentions of climate

change from its official websites and defund studies on global warming means a lot of those scientists do not want to put up with that bullshit. Typically, it takes months to even get a government job, and hurricane season is already here. It's been hanging out on the porch, most likely to storm through the door in the coming weeks.

Right now, as I record this, we've got an eye on Hurricane Erin, which, as of one hour ago from when I'm recording this, which is about six hours before the episode is released, Tropical Storm Erin could rapidly intensify in the coming days, they're saying. And it could be the first Atlantic hurricane of 2025. Stay tuned. But have those 450 positions been filled yet? No, they have not. So, the government fucked up weather-wise here; it would not be the first time.

So, in a moment, we'll talk about another gaffe that has become the stuff of legends. But first, let's donate to a cause of the ologist's choosing. This week, it's Kim's choice, and they selected the wonderful Trevor Project, which is the leading suicide prevention and crisis intervention nonprofit organization for LGBTQ+ young people. They provide information and support 24/7, all year round. And their website notes that the Trump administration eliminated the specialized suicide prevention support for LGBTQ youth callers through the 988 Suicide and Crisis Lifeline. The Trevor Project, however, provides immediate crisis support anywhere in the US via text, chat, or phone. The Trevor Project is 100% confidential and 100% free. You can find out more or donate to them at [TheTrevorProject.org](https://www.thetrevorproject.org). So, thank you to Kim for choosing them.

[Ad Break]

Okay, where were we? Yes, hurricane fuck ups.

Alie: How did meteorologists feel in 2019 with Dorian's map being altered?

Matt: [laughs] Sharpiegate!

Alie: What was going on? What was the reaction there?

Matt: Extreme frustration and outrage, to be quite honest. You know, it's just... It's so petty, it's so stupid, it's such a distraction from the actual concerns.

Aside: Okay, so this is officially known as "The Hurricane Dorian-Alabama controversy," and it involved a press conference in the Oval Office of then-president and now-president, Donald Trump, holding up a poster board map of Noah's predicted "cone of uncertainty," that's like, where the storm may hit. It was headed over Florida, Dorian was, and up the Atlantic coast. But someone had taken a black marker to the map and added a hump to the top of the cone to include Alabama. It was likely not a scientist, as all meteorological predictions at that point correctly stated that Alabama would not be hit. And the President, who tends to sign all of his papers with a black marker, later criticized a reporter for fact-checking him. Also, doctoring or altering official weather predictions for publication, or by a government official, is against the law, and it's punishable potentially by imprisonment. But so are a lot of things that aren't.

Now, what was the response like internally?

Matt: What I loved was the National Weather Service in Birmingham kind of stood up for... Really, their people, to be honest. Because they didn't want their people to be scared, and they didn't want a panic in their office. They're like, "No. This isn't going to happen. This is not a concern for us." And they did a great job of kind of rebuffing that and standing up and doing such a good job. But it's interesting you bring that up, because literally today, Andrew Freedman, who is a climate writer for CNN, he came out with an article that said that two of the NOAA agency employees that led the review into Sharpiegate and the discipline review were put on leave today. [Alie sighs] Yeah.

Alie: Really?

Matt: That's what we're up against right now.

Alie: [*sighs again*] Yikes. It's just... Yeah, it's so surreal.

I wanted to ask: When we're looking forward to, say, this season, how are you getting that idea of it's going to be a bad one, it's going to be a good one? And is it just getting worse every year with climate change?

Matt: [*laughs*] Yeah, so the climate change factor is just a kind of a layer of complication that's just added to everything. You know, that can be said for everything beyond hurricanes, too. But from a seasonal forecasting perspective, when you're sitting in the beginning of hurricane season, you're like, all right, how bad is this going to be? You're looking at what are water temperatures doing? Are they really warm? Are they really widespread warm? Are they just warm in a certain area? You want to get a sense of is the fuel going to be there to get these things going as the season goes on? In recent years, it's been kind of crazy, actually, how warm the oceans have been. Like, your initial thought is like, "Here we go again! Another hurricane season here, this is going to be bad."

And then you just do have to think about climate change as well and how that layers into everything. There's a whole body of research on climate change and hurricanes and how they're impacted, and it's complicated and frustrating and not great overall. So, you know, you're just trying to... I think we all pad a little bit on top of where things are just because of the world we live in today.

Alie: Yeah. Do you feel like it's not when you say it's not great? Do you mean the research isn't great, or it's not looking great? The prognosis is not great?

Matt: It's not looking great. [*laughs*]

Alie: [*laughs*] Yeah, okay.

Matt: Particularly... [*laughs*]

Alie: [*laughs*] I'm sorry. That's what I was afraid of. But just, like, gallows humor. It's so bad. Oh, my god. Okay, yeah. Continue.

Matt: I thrive on gallows humor. That's the only way you can in this industry, I think.

Alie: [*laughs*] It's like all you can do. The fuck, you know. So, like, all signs point to: It's not looking good.

Matt: Yeah, we're seeing more cases of storms that intensify rapidly, and it's happening right up to landfall now. It used to be, the conventional wisdom, particularly in the Gulf, was a storm would peak in intensity as it approached land, it would start to level off a little bit, and you'd have a bad storm. That's that. But what we've seen so many times in the last six, seven, eight years is these storms that just throw on the accelerator, they slam on it, and they never hit the brakes until they hit land. So, you're getting all these storms that are coming ashore at their maximum intensities. And that's why we've seen Category 5s, Category 4s just litter the whole Gulf Coast since 2017. The research kind of backs this up. This is not something that should be considered out of line going forward; this is going to become kind of more the rule than the exception, I think. [*Alie gasps*] And yeah, it's really, it's...

Alie: Argh! What the— Whaaa...?

Matt: Yeah. [*laughs*] You know, and it goes beyond that.

Alie: Well, if you are paying any attention to what's happening in the United States, for any listeners who might have blissfully been unaware, Emily Knudsen, Maddy Julian, both first-time question-

askers, wanted to know, Emily asked: How will cuts to the National Weather Service impact our ability to track hurricanes? Maddy wanted to know: How screwed are we this hurricane season because of the funding cuts to NOAA and the lack of satellite data? Allison Clark said: Hate to be a bum, but how bad are things expected to be in future storms as a result of these recent cuts to weather services?

So, I mean, part of why we're, like, laughing through tears is it couldn't have taken a worse turn here in terms of protections, understanding of it. Like, it seems surreal. What is going on with those programs? Is there any hope? Are there ways that scientists have been able to hold on to data or hold their positions or, like, what's going on inside? Is it as bad as the headlines? Is it not as bad as the headlines?

Matt: Yeah, it is. It just is. It's something. *[laughs]* I don't know what. No, but in all seriousness, like, you know, we're recording this in late July, and I mean, tomorrow, something could come out, some edict, and the whole thing changes. So, as it stands today, there have been a number of cuts throughout the course of the year. We had basically, like, not forced retirements, but heavily encouraged retirements, with scare quotes. We've had people that have left early. So, number one, you've lost a whole heck of a lot of experience from the National Weather Service. You've lost, for the young people in NWS, now they've lost mentorship and things like that, things that will help them to get better as time goes on. So, that's unfortunate, and you know, that ship sailed. So, that's done.

Because of the staffing cuts, they can't launch weather balloons as frequently in some parts of the country. That's a problem because then you lose some of the inputs to weather model data that are really important. And like, you think about it's 2025, we're still launching weather balloons, but the reality is that's one of the most critical inputs to weather models is getting this truth from the upper atmosphere. You know, you launch a weather balloon, you take all these readings, and all that stuff kind of gets ingested into models, and the models do better. They're based on what the atmosphere looks like in a 3D perspective, and that's what helps because no one lives 20,000 feet up.

The issue is, is that the proposed budget for 2026, I mean, it just took a machete to NOAA. And what people don't realize is that all this research is so critically important to our understanding of weather and our understanding of weather forecasting. Related to hurricanes, one of the proposed cuts was to eliminate the Hurricane Research Division, which is run in Miami, and *[Alie whimpers]* you do that and your forecasts aren't going to get any better. They may even regress. So...

Alie: Fuck, dude. Now? Like... *Now?* Of all times? To do that! Also, like, the President lives in South Florida most of the time, correct?

Mt: *[laughs]* Right.

Alie: That seems counterintuitive for...

Matt: It's counter-everything.

Aside: I asked disasterologist, Dr. Sam Montano, what is happening, and she sent me a note titled "Monday Morning Email of Doom." I will read it to you. So, Sam says:

The quick summary of what's happened is that the Trump administration has begun gutting FEMA with continued threats to eliminate the agency entirely. About 2,000 FEMA employees have left the agency or been fired since January. Perhaps most concerning, the agency is undergoing a "brain drain" with key, experienced leaders leaving in droves. The most recent is a guy who was in charge of FEMA's Urban Search and Rescue Teams.

The administration has taken away funding for state and local mitigation projects, preparedness efforts, and ongoing recovery efforts. I can't give you exact amounts because they're stopping funding in piecemeal ways, so it's almost impossible to track, but it's into the billions of dollars. There are ongoing lawsuits about it. Unclear how this will play out.

Trump has yet to formally nominate someone for the Senate to confirm to lead FEMA, so they put in this guy, David Richardson, as the temporary head. He is, and I cannot stress this enough, completely unqualified. He has zero emergency management experience, didn't know there was a hurricane season, and also has terrible fashion sense.

Now, Dr. Montano had hyperlinked those last words, and I couldn't click them fast enough. It led me to her July 2025 blog post, which informed me that Richardson didn't show up in Texas until nine days after the recent July floods, which have killed 135 people and counting.

"Incomprehensible," she says. Nine days. Nine days. Now, Sam's blog post also furnished photos of Richardson in collared shirts, unbuttoned to below his nipple line, and she offered the description of his arrival in Texas.

This man rolled in looking like some washed-up mobster coming off a three-day bender at the club. It's giving creepy, old, divorced dad at the bar, scamming on young women. This is the wardrobe of the sleazy guy, Olivia Benson [SVU's Mariska Hargitay] interviews when a socialite ends up dead. A public service ad for "cover your drink."

He was wearing what appears to be a Cartier watch and the dumbest hat I have ever seen. I know this is a tired exercise, but can you IMAGINE if [the former director] had shown up in Western North Carolina, over a week late, wearing a \$7,000 watch? Can you IMAGINE.

It might seem like I'm just being mean, bullying a grown man like this and all, but his fashion choice is actually indicative of all the problems with Richardson serving as the head of FEMA. (Think Miranda Priestley's Cerulean Sweater Monologue from The Devil Wears Prada.) His actions have demonstrated a complete disregard for the culture of the profession, a lack of respect for disaster victims, and an abdication of duty toward disaster survivors and emergency managers. He has once again demonstrated that he does not take his job seriously and has no interest in bettering FEMA. He is sloppy, unprepared, self-involved, a poor communicator, unperceptive, and ignorant.

Okay, again, Sam is a professional disasterologist who has been doing this much longer than David Richardson. Okay, back to Sam's email, "Monday Morning Email of Doom," again, talking about current senior official performing the duties of FEMA administrator at FEMA, David Richardson.

The biggest problem is that, perhaps because FEMA is under the Department of Homeland Security, Kristi Noem has effectively taken the agency over. This is causing all kinds of ethical problems and bureaucratic hurdles. The Trump administration's plan (which aligns with Project 2025) is that states are going to need to be responsible for their own disasters. The problem, of course, is that states do not have the people, budget, or expertise to be able to do that effectively for big disasters and catastrophes. This is probably much more than you're interested in...

No, Dr. Sam Montano, it is not.

... But there was a big new nationwide survey of local, state, and territorial emergency management agencies. The TLDR is that there are not anywhere near enough people and funding for local, state, and emergency managers to meet the needs of their communities. What that means is emergency managers can't do enough to try to prevent disasters and to

prepare for them. So, when those disasters inevitably happen, the response and the recovery won't go well.

In italics, Dr. Montano stressed to me:

We are not there yet. But if these federal changes continue and state and local governments don't dramatically increase funding, we are on a trajectory where public sector emergency management will collapse.

So, what should people do? Sam says:

1) Call your reps. If you live in America, tell them to demand that the administration nominate a qualified FEMA administrator. Urge your reps to pass legislation to make FEMA independent from the Department of Homeland Security.

Also, for anyone who doesn't even live in the United States right now, 2) Take as many individual actions to prepare yourself for disaster with the resources that you have. The usual things like stockpiling extra food, water, medications, having a family communication and evacuation plan, money set aside for an emergency if you can. "Buy flood insurance," Sam says. "Please, please, please buy flood insurance."

Also, be proactive about getting disaster alerts and warnings. Make sure your phone settings allow for government emergency alerts (this is the screeching sound your phone makes when there's a tornado, etc.), which can get a bit annoying, but you really need to keep it turned on. You should also Google your city or town or county or country and "emergency management" and see what other emergency alerts you can opt into. You can also follow your local emergency management agency on social media.

4) Check in with your friends and family and neighbors and make sure they've done all those things, too. Especially, consider grandparents, elderly neighbors, or other people in your life who might need a little extra help during a disaster.

5) Prepare for federal assistance to be delayed or much less than in the past, and advocate for community-wide preparedness where you live. "Look into your local emergency management agency's budget. I bet they do not have enough money. So go to a town council meeting and advocate for increasing their budget, or taking on more local mitigation projects." Also, you can urge local climate groups to include disaster advocacy in their climate work.

So, we're going to link all those actions and advice via our website at AlieWard.com/Ologies/Tempestology, which will be also linked in the show notes. And also, check out Dr. Montano's Disasterology episode and follow her on Bluesky and Instagram @SamLMontano. She is a force for good.

More good...

Matt: Now, here's the good news. The good news is that Congress, sometimes, they can be the grownups in the room. And weather, generally, is a very bipartisan issue; weather affects everybody. They don't care who you voted for; everybody in their district doesn't want to be accused of cutting, you know, something that impacts weather forecasts that leads to a tragedy in their town. So, the Senate came together and said, "No, we're not going to do these cuts. Sorry. In fact, we're going to actually increase funding to NOAA a little bit, and we're going to keep all these agencies that you want to eliminate, all these research agencies that you want to eliminate, we're going to keep them all." Okay, good. But the House actually came back and said, yeah, actually, we're going to increase the funding.

So, the hope here is that eventually, when all is said and done, this will be a moot point. The funding levels will hopefully be at least the same as they've been, if not a little bit higher, and, you know, all is hunky dory and well with the world. But hope is not always a strategy. It even remains to be seen whether or not, you know, the administration even wants to play by the rules and do what Congress tasked them with.

So, we're living in an interesting time. I don't want to say we're screwed because I think that that's a little strong. But suffice to say, I and most other meteorologists I know, we're all concerned, deeply concerned, and troubled by what's been happening.

Aside: And funding these science agencies, even if they're working on climate science, behooves everyone.

Matt: If you are the most fiscally disciplined individual and you believe that we should be very, very fiscally conservative as a country, that's fine. But you're saving literally like next to nothing. But the cost that you're incurring because of that is significant! You know, you're risking... You're playing with fire, literally and figuratively, and you don't want to do that.

Aside: And the cost, other than human life? So, from FEMA's own website, based on 2018 analyses, "natural hazard mitigation saves \$6 on average for every \$1 spent." And remember that this was a 2018 study, and numbers have trended upwards when it comes to savings. FEMA's report also acknowledged the social and the psychological downstream effects of disaster and noted that:

Some mitigation benefits include the reduction in domestic violence, the conservation of heirlooms and photos, and the preservation of community and culture. Those benefits can be extremely difficult to quantify, [and so], they were omitted from the analyses.

So, that six-to-one is definitely a conservative estimate.

Alie: Like, if every hurricane had a price tag so it actually hit home with people, you know? Like, "Oh, this is how much this hurricane is going to cost," people might care more than just categories, or even lives, which is astonishing and horrifying. But I think we've become, especially in America, we have become so used to mass casualties in a way that's just pretty horrifying. But one thing that America still runs on is money. It's interesting to think damage-wise, what that might do. [*"It is the international system of currency which determines the totality of life on this planet."*]

Should you go by the Waffle House index, or no?

Matt: [*laughs*] The Waffle House index is undefeated. You have to go by that, right? [*laughs*] When you're making your mental forecasting ensemble of how things are going to go, you have to use the Waffle House as an input. No question about it.

Aside: So, the Waffle House index refers to the 365, 24/7 purveyors of breakfast and diner foods that dot the Midwest and the southern United States landscapes. It was kind of a folk guideline that if a nearby Waffle House is still open, you're probably safe because they close for no man. It would be amazing if FEMA used this on-the-ground metric.

But wait, they do, or they did, under the direction of Obama-appointed William Craig Fugate, who told the *New York Times* in 2012 that, "Waffle House has a very simple operational philosophy. Get open. They never close. They run 24 hours a day." So, if the local Waffle House is business as usual, things are fine. Now, if they have a limited menu, their power might be iffy, and if they have shuttered the doors for a storm, you better get your ass to shelter, too. That is the Waffle House index. Now, the *New York Times* article also laments that "After its poor handling of Hurricane

Katrina in 2005, FEMA was the Homer Simpson of federal agencies, a symbol of pitiful incompetence." It's a good thing that Katrina did not hit a nuclear reactor.

Alie: I know that there's so much that is challenging about being a tempestologist in the year of our Lord 2025, [*sighs*] the year of our Beyoncé 2025. [*Matt laughs*] What's the hardest thing about it?

Matt: Oh, that's a good question. The hardest thing, what I'm learning, is effectively reaching people. You think you've got a method of communicating with people. But, you know, as a scientist, sometimes you assume people know more than they do. Sometimes you expect people to know more than they do, and you've just got to be so aware and cognizant of that. So, it's not just even thinking about the forecast and the weather and the science. It's thinking about the lingo that you use to convey something, because one word can be very different than another. We are constantly learning, constantly learning how to deal with it, and communicate to people. It's a fun process, but it's hard. It's hard, and you feel a responsibility to do it right. So, anytime you come up short in that or something gets misinterpreted, you feel kind of sucky, you know? Even if you did the best you could, but still.

Aside: What about faves? Let's start with Kim.

Alie: What do you love about them? Or working in hurricanes?

Kim: I have learned so much about nature, about geography, about engineering, about coding, about visualization, because I study hurricanes. I have connected with all kinds of people because I study hurricanes. And the communities that build around these shared values of investigating fascinating phenomena and distilling that information in ways people can actually act on to make their lives better, like, the fact that I'm talking to you right now is an example of one of my favorite things about hurricanes. [*Alie laughs*] Like, 20 years ago me couldn't even fathom that I would, you know, publicly speak in any way, shape or form, especially recorded. People could listen to this again? Like, what? But just sharing science is so valuable because it's connecting humans to other humans over things that we don't have control over, but by learning more about them, we can better control what we do in the face of them.

Alie: That's such a beautiful point.

Aside: Any advice for someone who thinks that they might want to go into hurricane research or storm research? Like, they either want to chase a storm on radar, or in an aircraft shaking violently with turbulence and fear?

Kim: This might be the weirdest thing, but learn your way around a computer. It's amazing how much data are out there that you could potentially put your own perspective on if you have a little bit of knowledge of computers, of accessing data, and programming languages like Python that are open source. Like, all the visualizations I do are made possible because programs like Python are just... People just post their code to Stack Overflow or GitHub, and they're like, "Here, take it." And so, I've started doing the same thing for other people. Equipping yourself with a skill in pursuit of something you're passionate about, that skill can be used in other ways too, but you also have more fun learning it (because learning is hard) if you're doing it in pursuit of this really interesting thing. And being able to visualize data, interrogate data, all that kind of thing, it can serve you well in so many ways, and also just give you insight into how to interpret what other people are sharing.

Alie: Oh, that's so smart. I definitely wouldn't have known how to audio edit if it weren't for *Ologies*. I was like, "I don't want to learn that." But then I was like, "I do want to make a podcast. I'm okay with it." [*laughs*] That's great.

Aside: And what does Matt adore about these storms?

Alie: Is there a favorite thing you have about studying hurricanes, about being a meteorologist, about what you do?

Matt: I just find it all so fascinating. Hurricane Harvey changed me a lot because it was the first time that I had dealt with, you know, a forecast that I had made, and it was for a devastating event. So many people I knew were impacted by it personally, that it was a little hard to reconcile. As a weather geek, a lot of times you can sit and watch these storms from afar and you're like, "Yeah! That's awesome! That's so cool!" because it's nature. You know, the reality is it impacts people really hard. So, that really kind of muted some of my enthusiasm, stuck the air out of the balloon a little bit, but I still am fascinated by it.

You learn how to compartmentalize all your emotions a little bit. You go back to just appreciating the science for what it is. And it's just, the power of hurricanes is incredible. They are fascinating. They are so impactful, not just at a human scale, but like we've talked about, just at every scale imaginable on the planet. They're just the most fascinating things that exist. It emphasizes the power of nature over man, and just constantly, you're learning something new and seeing something incredible.

And I mean, even honestly, with climate change, it sucks, it's horrible, it's awful. But I'm here now, I'm doing this now, and it's a responsibility to try and understand everything better because it is going to get worse and, you know, it's an opportunity for you to do, to ultimately try and do good for people in a world that is not necessarily doing them good. At the end of the day, if you could say that you had a positive impact on helping someone prepare for and get through a storm, then great. That's awesome. I feel like I did something good.

So, once again, ask delightful people, disastrous questions. And thank you both so much, tempestologists Matt and Kim, for being two of the finest ever. You can follow them at their social media handles, Blue Sky, @MattLanza, Kim is @DrKimWood, which are linked in the show notes or our website at AlieWard.com/Ologies/Tempestology.

We are @Ologies on Bluesky and Instagram. I'm @AlieWard on both. *Smologies* are those shorter, classroom-safe versions of *Ologies* classics, those are available wherever you get podcasts. And you can sign up at Patreon.com/Ologies to send in your questions ahead of the recording. Erin Talbert administers the *Ologies* Podcast Facebook group. Aveline Malek makes our professional transcripts. Kelly R Dwyer does the website. Noel Dilworth tracks our unpredictable schedule. Susan Hale oversees it all as managing director. And pilots of our always exciting and shaky flight to hit publish, are Jake Chaffee and Captain Mercedes Maitland of Maitland Audio. Nick Thorburn wrote the theme music. Extra help provided by Jarrett Sleeper of MindJam Media, who's also a wonderful husband.

And if you stick around till the end of the episode, I burden you with a secret from my heart. This week, it's that I realized I'm always afraid when I'm in stores that people will think I'm shoplifting. The only time I've ever shoplifted once is right out of college when my blood sugar was really low and the line at Rite Aid was so long I was going to be late to work. So, I stole a Clif Bar and then I bought one later at Trader Joe's, and I returned the same Clif Bar to Rite Aid the next day because I'm an individual who is ruled by fear.

But nonetheless, I'm also just afraid in general that people might think that I would steal from them. And I realized, I was at the public library, and I realized it's so freeing because you can look at stuff without worrying about seeming weird or not buying anything. And then I was like, how fitting that *libre* means 'free.' Although, I just went down an etymology hole, and I was like, *libre*, free library, free borrowing. Also, a freedom of the soul. But it turns out that library comes from a root word

about tree bark, and it's not related to liberty at all. Still, libraries, a place to experience liberty liberally. And librarians, I salute you, and I'm not going to steal any books.

Okay, stay safe, please. And if you didn't like that this got political, I didn't like that science gets political either... I don't like it. They should fund science based on the science that we need and not based on appeasing coal companies and stuff. But what do I know? Enough not to mark a map with a Sharpie. Berbye!

Links to things we discussed:

[How Eric Berger and Matt Lanza Became Houston's Most Trusted Meteorologists](#)

[Hurricane Science Was Great While It Lasted - The Atlantic](#)

[FEMA: Mitigation Saves: Federal Mitigation Grants Save \\$6 per \\$1 Spent](#)

[What is the difference between a hurricane and a typhoon?](#)

[Websites hosting major US climate reports taken down](#)

[The Hurricane Hunters That Never Returned](#)

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[TWISTER Clip - "Cows" \(1996\) Bill Paxton + Helen Hunt](#)

[June 2025: Fired, rehired and fired again, some NOAA employees get letters demanding money](#)

[July 2025: Cuts to NOAA increase the risk of deadly weather tragedies](#)

[July 2025: Congressional committees push back on Trump administration's proposed NOAA budget cuts](#)

[Herbert Seymour Saffir's *Morro Castle* Survival](#)

[Severe Weather Topics - Wind Shear](#)

[NOAA's National Weather Service - Glossary - shear](#)

[25th Anniversary of a 'hairy hop' into Hurricane Hugo - NOAA's Atlantic Oceanographic and Meteorological Laboratory](#)

[The Wildest Ride on a Hurricane Hunter Aircraft - Eos](#)

[Structure of the Eye and Eyewall of Hurricane Hugo \(1989\)](#)

[Clement Lindley Wragge: The Life of a Pioneer Meteorologist](#)

['Inclement' Wragge : pioneer weather forecaster](#)

[Deaths Related To Hurricane Rita And Mass Evacuation"](#)

[Encyclopedia of Hurricanes, Typhoons, and Cyclones, New Edition](#)

[A review of ocean-atmosphere interactions during tropical cyclones in the north Indian Ocean](#)

[Natural Hazard Mitigation Saves Interim Report](#)

[Man Behind FEMA's Makeover Built Philosophy on Preparation](#)

Per Dr. Sam Montano:

['Setting us up for catastrophe': alarm at Trump attack on federal disaster agency: 1/23/25](#)

[Trump cuts will lead to more deaths in disasters, expert warns: 'It is really scary': 5/5/25](#)

[Deadly floods could be new normal as Trump guts federal agencies, experts warn: 7/8/25](#)

David Richardson's "[terrible fashion sense](#)"

How can people prepare for disasters?

1) Call your reps and tell them to demand that the administration nominate a qualified FEMA administrator, require FEMA to spend its funding as Congress has instructed, and pass legislation to [make FEMA independent from DHS](#).

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4) Check in with your friends, family, and neighbors and make sure they've all done these things too. Especially consider grandparents, elderly neighbors, or other people in your life who might need some extra help during a disaster.

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